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Google Data Analytics Capstone
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Case Study 1: How Does A Bike-Share Navigate Speedy Success?

Intro: This case study explores how Cyclistic, a Chicago-based bike-sharing company, can convert casual riders into members. Using 12 months of data, I've cleaned, analyzed, and visualized usage patterns to identify differences in riding behavior between rider types. My findings highlight how ride duration, day-of-week trends, and bike preferences can help marketing strategies to increase memberships.

1. How do annual members and casual riders use Cyclistic bikes differently?
2. Why would casual riders buy Cyclistic annual memberships?
3. How can Cyclistic use digital media to influence casual riders to become members?

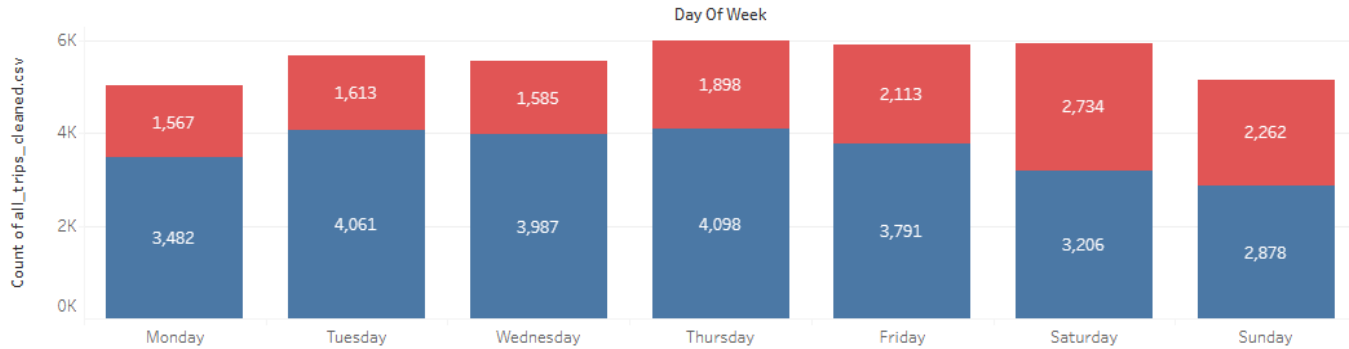
Problem: Cyclistic wants to understand the difference between casual and member riders so they can convert casual riders into members. They want to know what sets them apart and how they can use media to influence casual riders to become members.

Summary: I've combined 12 months of ride data from multiple CSV files, removed duplicates and errors, added calculated values such as ride length and day of week, and filtered out invalid data. After cleaning, I analyzed the data to uncover patterns in rider behavior and exported it to Tableau for visualization. The dashboards display weekly ride patterns, ride counts, average ride durations, and bike-type usage, helping reveal how casual and member riders differ in their habits.

Weekly Ride Patterns

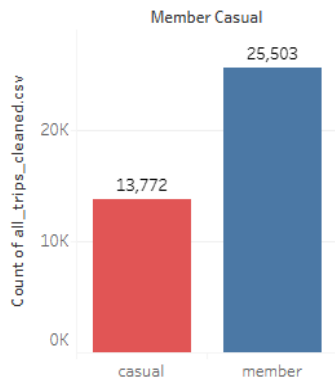
Member Casual
casual
member

Members ride all week consistently; casual riders peak on weekends.



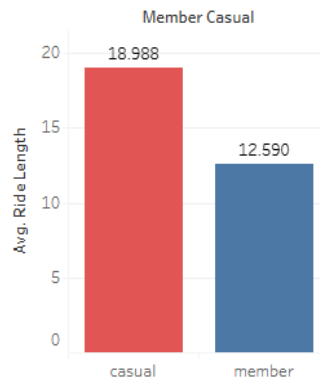
Ride Count

Members ride more than casual riders.



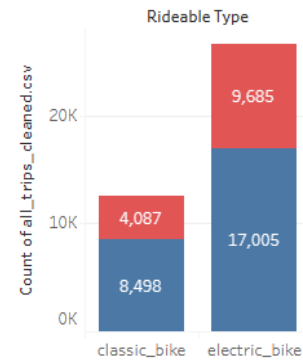
Average Ride Length

Casual riders take longer rides.



Rideable Type Usage

Electric bikes are preferred for casual use. Members like classic bikes.



Insights: Member riders maintain consistent activity throughout the week, while casual riders peak on weekends. Members use Cyclistic bikes for regular commuting, while casual riders use them mainly for leisure or weekend riding. Members take significantly more rides overall than casual riders, indicating stronger engagement and commuting use. Casual riders tend to take longer rides than members since they use bikes for sightseeing. Casual riders prefer electric bikes for ease and comfort, while members favor classic bikes for practicality and cost efficiency.

Key Findings:

- Members ride consistently throughout the week, and casual riders ride longer during the weekends.
- Casual riders take longer rides than members on average.
- Both parties prefer electric bikes, but members are more likely to ride classic bikes.
- These patterns suggest opportunities for targeted promotions and membership incentives.

Recommendations:

- Promote weekend membership deals to attract casual riders.
- Highlight the benefits of classic bikes and their convenience to frequent commuters.
- Use digital campaigns focused on converting long-ride casual riders into members.

Conclusion: Through this analysis, I explored Cyclistic's ride data in depth by using Google Sheets, BigQuery, and Tableau to visualize patterns in rider behavior. By examining weekly ride trends, ride counts, average ride lengths, and bike-type performances, I gained a clear understanding of how casual and member riders differ in their usage. This process helped me strengthen my data-visualization and interpretation skills while uncovering actionable insights that can guide Cyclistic's marketing strategies. Overall, this project showed how data-driven storytelling can reveal opportunities to convert casual riders into loyal members and support Cyclistic's continued growth.